

Problem

A voltage of $45e^{-2000t}$ V appears across a parallel combination of a 100-mF capacitor and a 12- Ω resistor. Calculate the power absorbed by the parallel combination.

Step-by-step solution

Step 1 of 4

The current through the capacitor is,

$$i_c(t) = C \frac{dv(t)}{dt}$$

Substitute $45e^{-2000t}$ V for $v(t)$ and 100-mF for C .

$$\begin{aligned} i_c(t) &= 100 \times 10^{-3} \frac{d}{dt} (45e^{-2000t}) \\ &= 100 \times 10^{-3} \times 45 (-2000e^{-2000t}) \\ &= -9000e^{-2000t} \text{ A} \end{aligned}$$

Step 2 of 4

The current through the resistor is,

$$i_R(t) = \frac{v(t)}{R}$$

Substitute $45e^{-2000t}$ V for $v(t)$ and 12 Ω for R .

$$\begin{aligned} i_R(t) &= \frac{45e^{-2000t}}{12} \\ &= 3.75e^{-2000t} \text{ A} \end{aligned}$$

Step 3 of 4

The total current through the parallel combination of the resistor and the capacitor is,

$$i(t) = i_R(t) + i_c(t)$$

Substitute $3.75e^{-2000t}$ A for $i_R(t)$ and $-9000e^{-2000t}$ A for $i_c(t)$.

$$\begin{aligned} i(t) &= -9000e^{-2000t} + 3.75e^{-2000t} \\ &= -8996.25e^{-2000t} \text{ A} \end{aligned}$$

Step 4 of 4

The power absorbed by the parallel combination is,

$$p(t) = v(t)i(t)$$

Substitute $45e^{-2000t}$ V for $v(t)$ and $-8996.25e^{-2000t}$ A for $i(t)$.

$$\begin{aligned} p(t) &= (45e^{-2000t})(-8996.25e^{-2000t}) \\ &= -404831.25e^{-4000t} \text{ W} \\ &= -404.831e^{-4000t} \text{ kW} \end{aligned}$$

Thus, the power absorbed by the parallel combination of the resistor and the capacitor is

$$\boxed{-404.831e^{-4000t} \text{ kW}}.$$